SUSTAINABILITY

Metrics of energy efficiency from servers to data centers, and DCEM workstream update

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OCP BRONZE Member

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Motivation

- 15 US states have datacenter moratoriums
- Datacenter build dilemma for peak power vs energy efficiency
- Energy cost vs system cost (memory/system price increase YoY)
- Metrics: build CPUs for Perf/W and Perf/mm2

- DCEM: create design metrics that cross the computing stack to evaluate efficiency
  - Minimum: optimization metrics
  - Ideal: design metrics
  - Top down metrics (HW/infra) vs bottom's up metrics (SW/Perf)

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Datacenter Trends

Liquid Cooling	High Ambient Temperature Operation
Improved cooling efficiency	Reduced facility cooling power
Additional non-IT power cooling overheads	Additional per-server fan power to compensate

PUE does not break down power enough to understand these trends

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Breaking Down Server Power

$$PUE = \frac{E_{Total}}{E_{IT}}$$

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Breaking Down Server Power

Infrastructure Utilization Efficiency (IUE)

$$IUE_P = \frac{P_{Total}}{P_{PSU\ Loss} + P_{Fans} + P_{Idle} + P_{Compute}}$$

- PSU Loss: quantify the PSU's efficiency converting AC to DC
- Fans: quantify per-server cooling power
- Idle: quantify the baseline power for keeping a server on
- Compute: quantify the power of running workloads

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Power Breakdown Methodology

Estimate component power from server-level measurements

PSU Loss	Reproduce 80 PLUS PSU efficiency curves at varying loads
Fans	Fan power $\propto$ (fan speed)^3, monitor IPMI
Idle	Idle the server and discount PSU loss and fan power
Compute	Run workloads, measure server power, discount the rest

Check out HotInfra 2026 paper for measurement methodology details

August Ning, Xavier Ouvrard, Babak Falsafi,
"Power Breakdowns and PUE Implications of Air-Cooled and Direct-Liquid-Cooled Servers"
HotInfra 2026

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Hyperscaler vs Co-location Operation

Case Study Assumptions

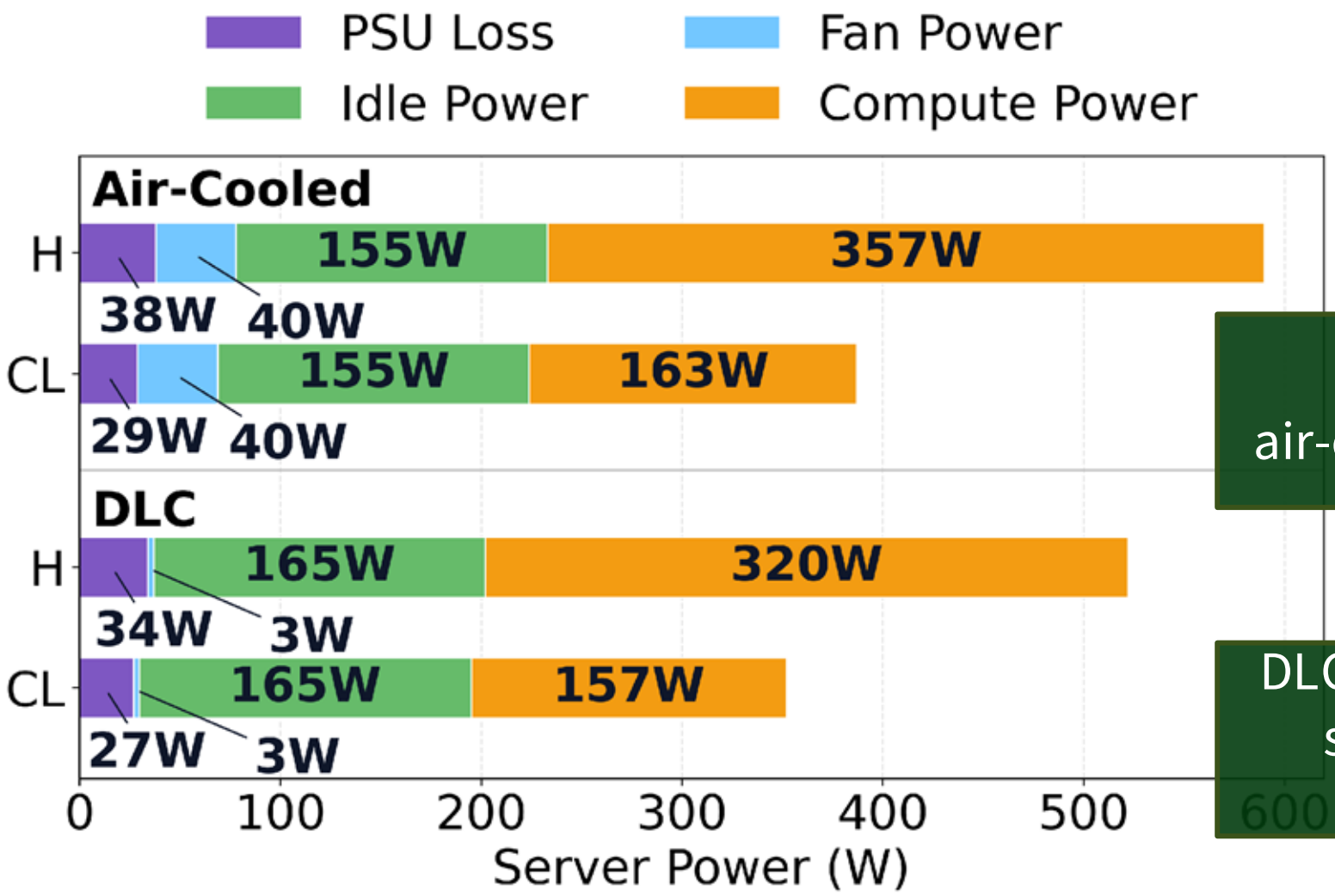
Use Case	PUE* (24°C)	Server Utilization	DLC Fan Speed	DLC CDU Power†
Hyperscaler	1.1	60%	20%	20W
Co-location	1.3	15%	20%	20W

*Assuming identical racks and cooling solutions throughout the datacenter, negligible PDU loss
†Assuming 200W CDU pump power for a 20kW rack with 10 servers

Babak Falsafi, “To Measure is To Know: Breaking Down Datacenter Power Consumption”, Computer Architecture Today, 2025
“Custom Cooling Fan Options for Dell EMC PowerEdge Servers”, DELL EMC Technical White Paper, 2019
Sebastian Moss, ”Meta ups server room temperatures to 90 degrees F, in effort to reduce water usage” Data Center Dynamics, 2022

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High Ambient Temp. Operation



High ambient temp (32°C) requires increased fan speeds DLC server CPU cooling is independent of ambient temps

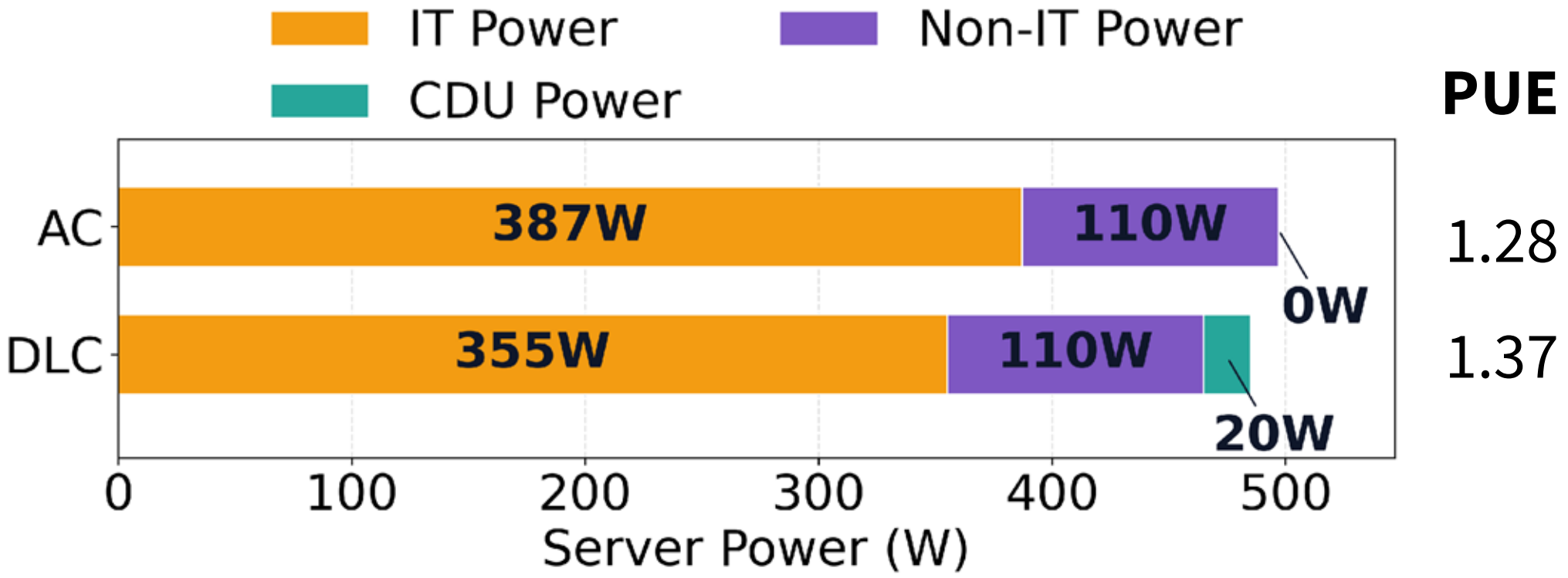
Fans consume 10.2% of air-cooled co-location server power

DLC saves 9.0% (CL) to 11.5% (H) of server power compared to air cooling

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Air-Cooled vs DLC PUE @ 32°C

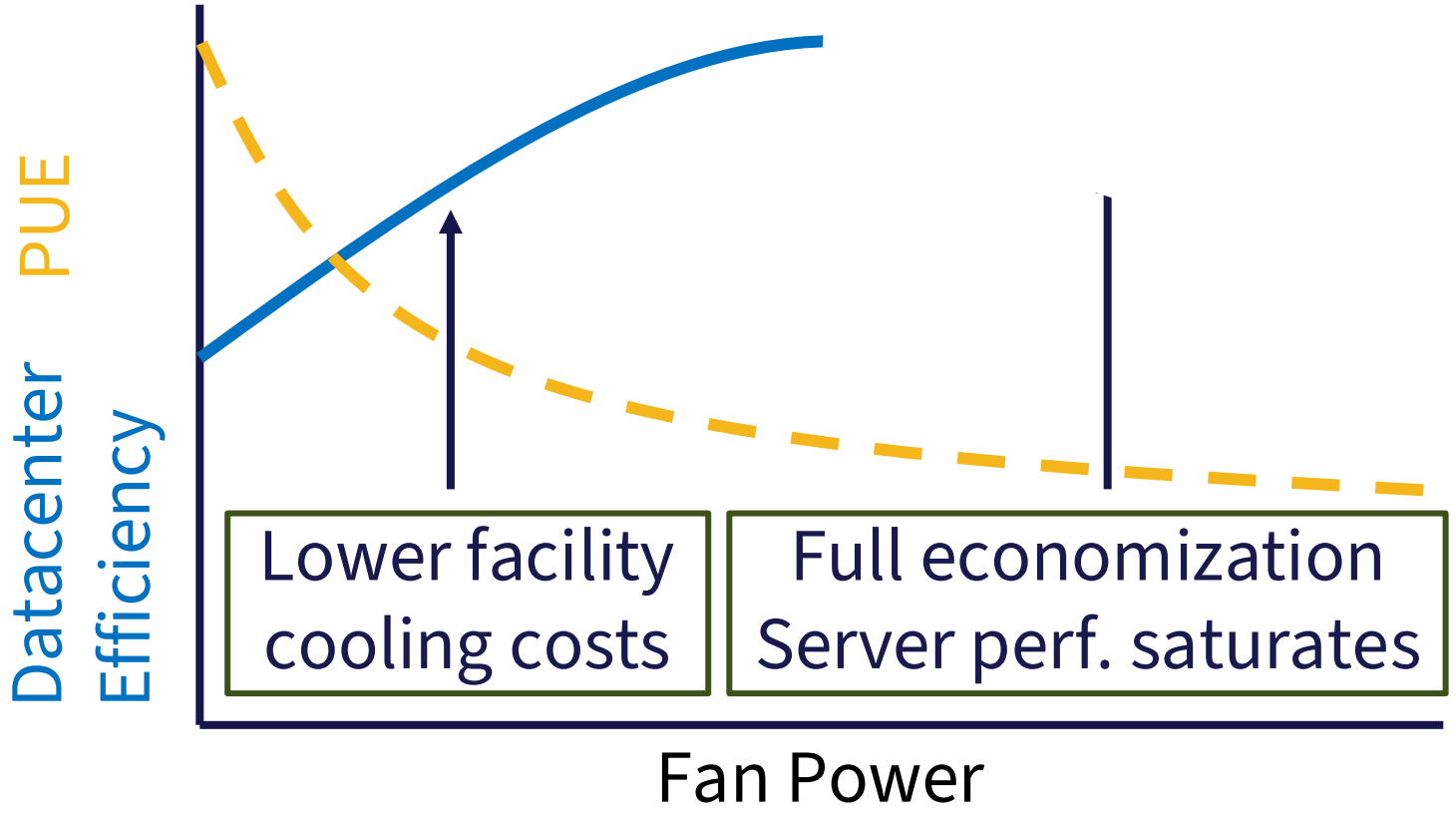
DLC datacenters use less power per server,
but have additional facility power overheads



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Fan Power and PUE

Server fan power is considered IT power →
increasing fan power decreases (improves) PUE!

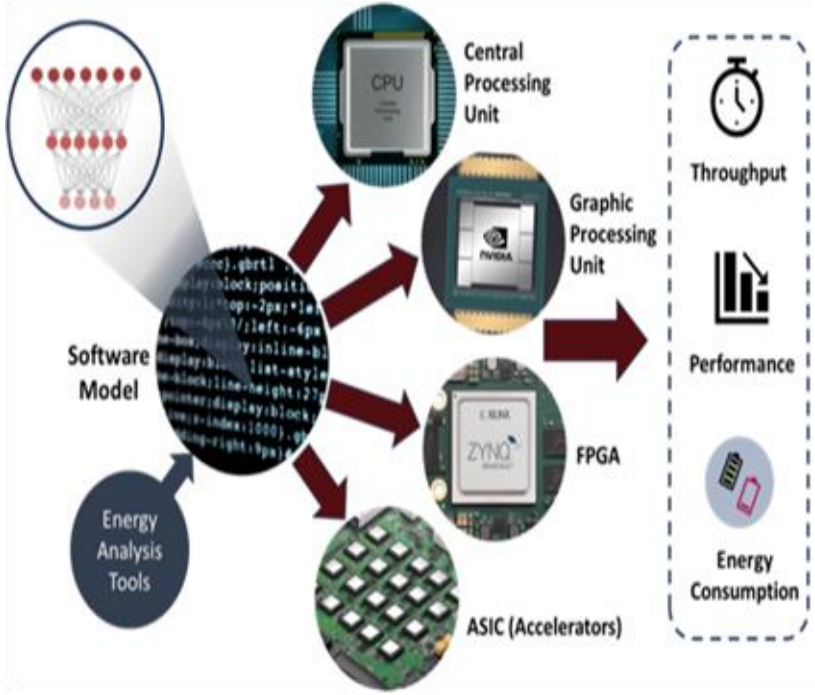


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Measurements from Components to Data Centers:

Metrics for the Data Centers

- **Environmental and Physical**
  - Temperature: CPU, GPU, RAM, Fans, etc.
  - Airflow
- **Electrical and Power**
  - Power (Instantaneous): CPU, GPU, RAM, Fans, etc.
  - Energy (Cumulative and Interval): CPU, GPU, RAM, Fans, etc.
- **Operational and System Metrics**
  - Runtime: Execution Time, Latency
  - Utilization: Core usage, memory bandwidth saturation, PCIe throughput
  - Clock Speeds: Cores, Memory, Fabric
- **Computational and Derived Metrics**
  - Throughput: FLOPs, GFLOPS, Ops/second
  - Efficiency Ratios: GLFOPS/Watt, J/Ops, J/bit, J/liter or Kg of cooling, J of Compute/J of Cooling
- **Other Metrics (in development)**
  - Algorithms/Software, Architecture, Hardware

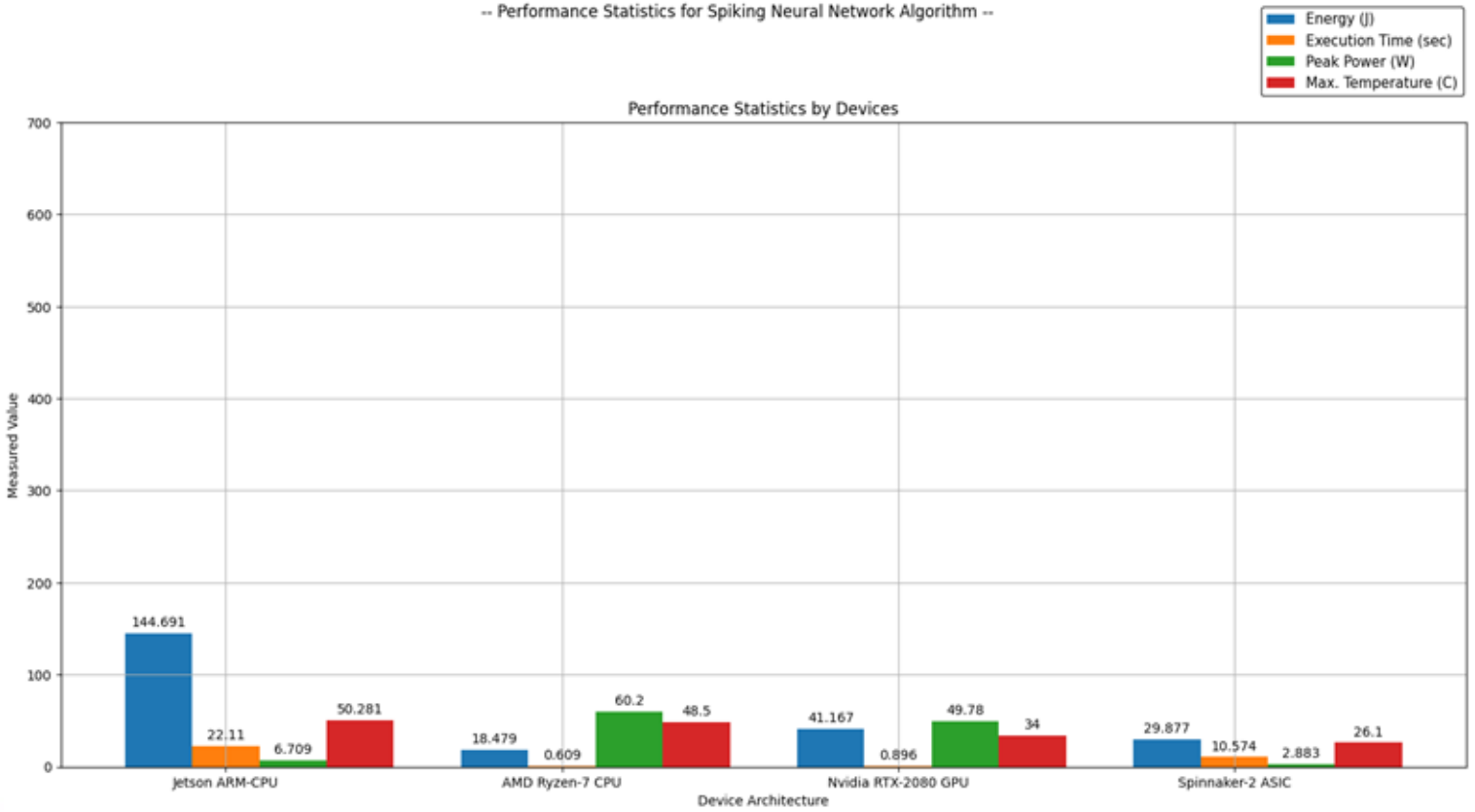


A. Bhattacharjee, C. Scrosati, M. Ramos, Z. Hu, & *S. Shankar*

OCP GLOBAL SUMMIT 2026 Scaling Innovation for the AI Era.
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Measurements: Spike Neural Networks

Device	Energy (Joules)	Time of Execution (sec)	Peak Power (W)	Max. Temperature (°C)
Jetson ARM-CPU	144.691	22.211	6.709	50.281
AMD Ryzen 7	**18.479**	**0.609**	60.2	48.5
Nvidia RTX 2080 GPU	41.167	0.896	49.78	34.0
Neuro ASIC	29.877	10.574	**2.883**	**26.1**



Ack: H. Gonzalez, J. Hönig, C. Mayr

- **Neuromorphic-specialized architecture (ASIC) is energy efficient and thermally cooler for SNN Algorithm**
- **Although CPUs seem more efficient than GPUs, the time of execution makes them less energy efficient over the simulation**

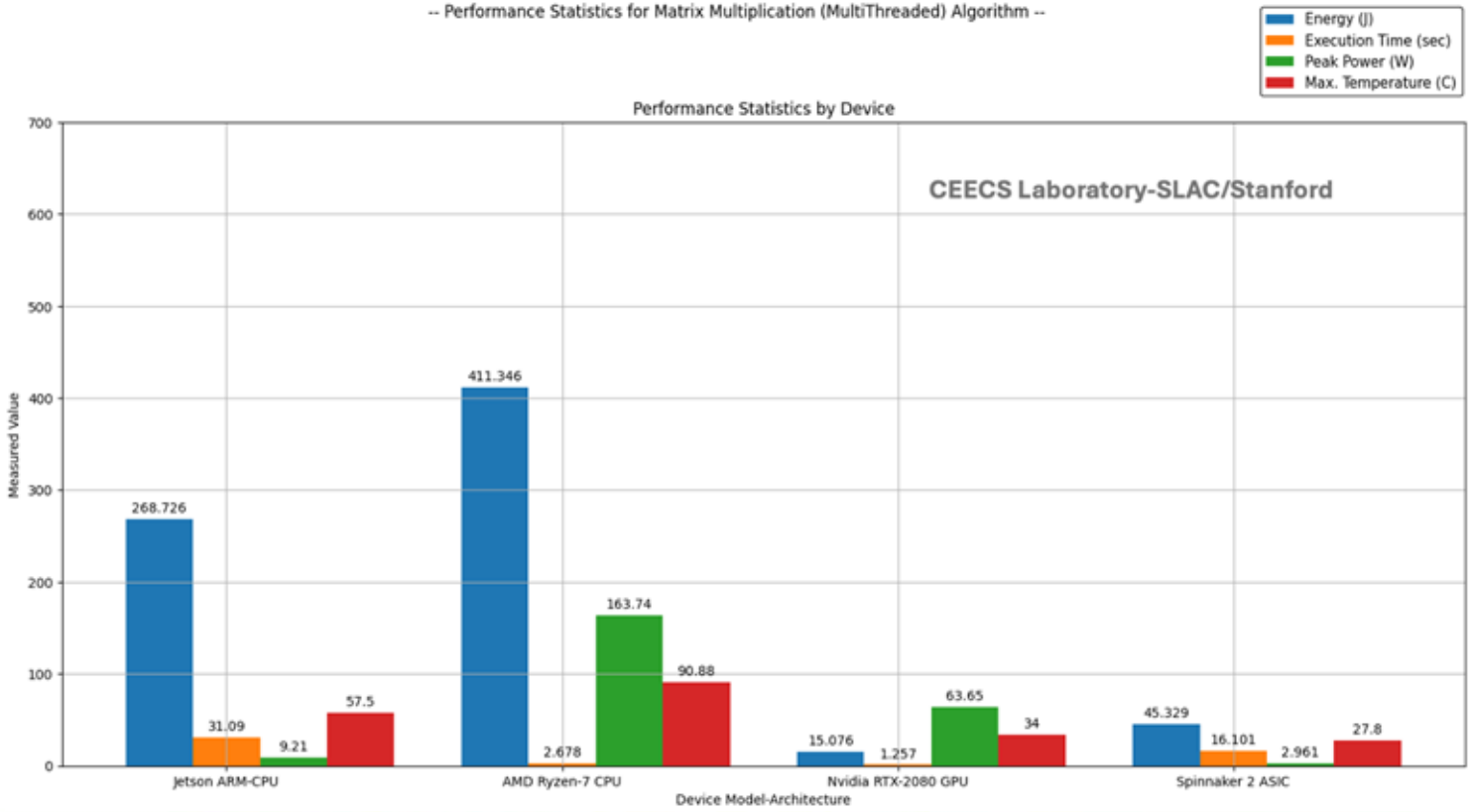
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Measurements: Matrix Multiplication (Multithreading)

Device	Energy (Joules)	Time of Execution (sec)	Peak Power (W)	Max. Temperature (°C)
Jetson ARM-CPU	268.726	31.09	9.21	57.5
AMD Ryzen 7	411.346	2.678	163.74	90.88
Nvidia RTX 2080 GPU	**15.076**	**1.257**	63.65	34.0
Neuro ASIC	45.329	16.1009	**2.961**	**27.8**

**GPU, ASIC data same for both Naïve and threaded cases.*



Ack: H. Gonzalez, J. Hönig, C. Mayr

- **GPUs appear more energy efficient for Matrix Multiplication, for which the architecture appears well-optimized**
- **Neuromorphic-specialized architecture (ASIC) is less energy efficient for Matrix Multiplication Algorithm than for SNN**

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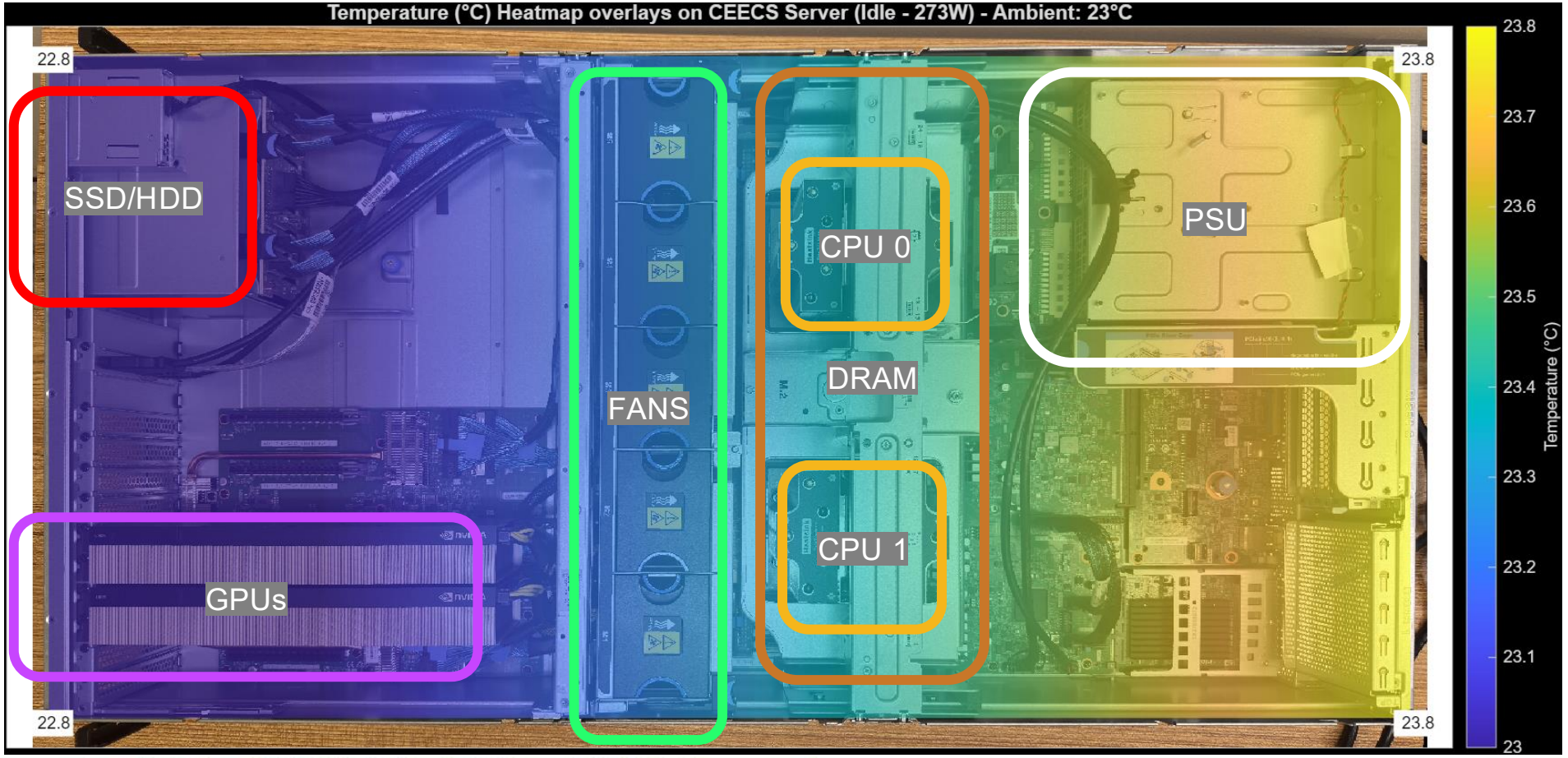
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Temperature Measurement Methodology

- Device measurements can be performed automatically and manually
  - Manually the HUATO S220-T8 device provides a sensing interface for K-type thermocouples.
    - For the following example, five thermocouples are used to take measurement
  - Automated Measurement
    - CEECS Device
      - External Sensing Device: To automate external measurement, the CEECS measurement device utilizes MCP9601 Thermocouple amplifiers in addition to K-type thermocouples to measure specific points in a system. A LPS28DFW Temperature and static pressure Sensor is used for airflow and temperature sensing
      - SNMP Interface: If present in server hardware, SNMP can be used for continuous remote temperature monitoring

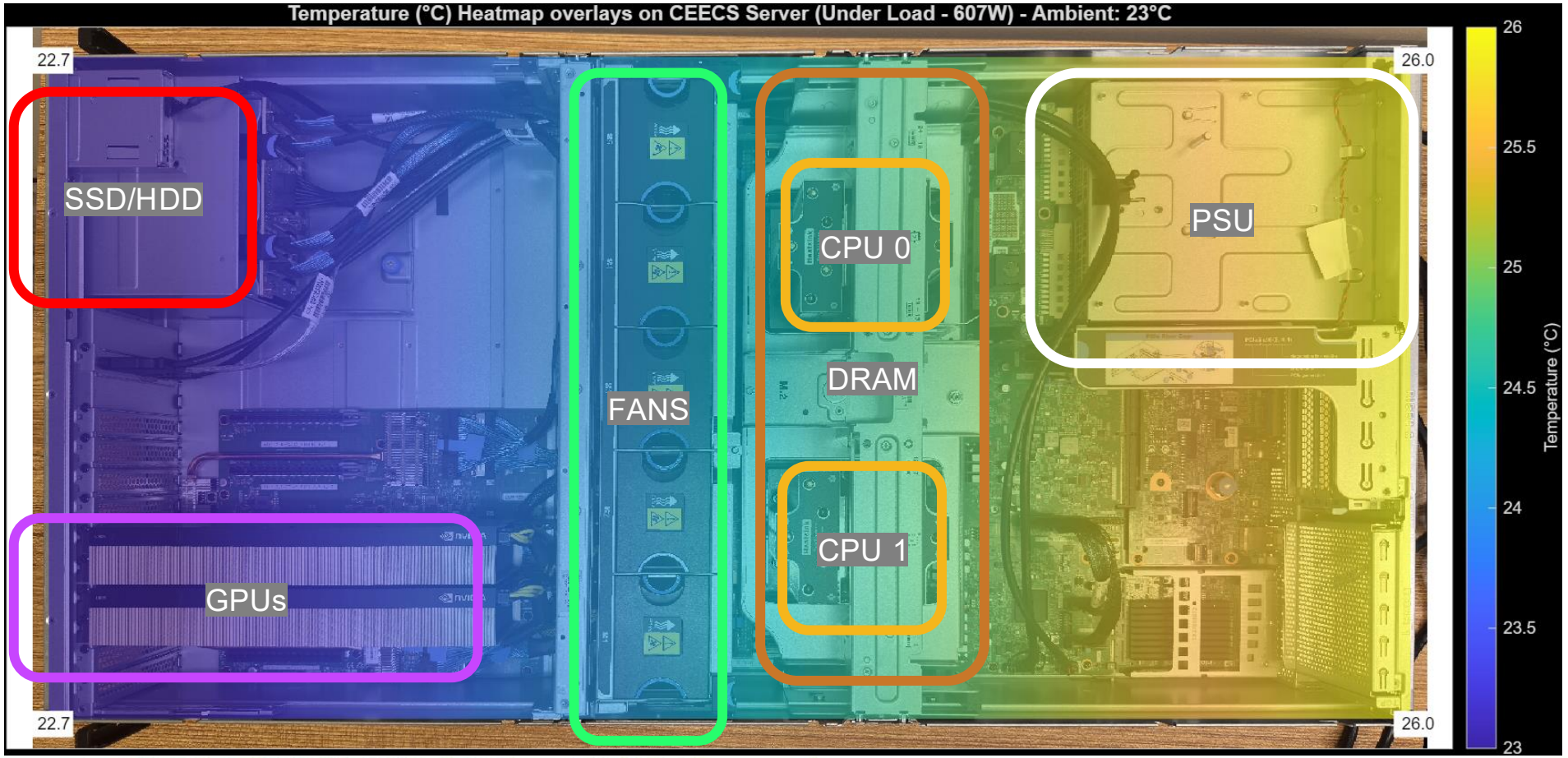
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Manual Measurement:
Lenovo SR675 – Temperature Heatmap (273 W)



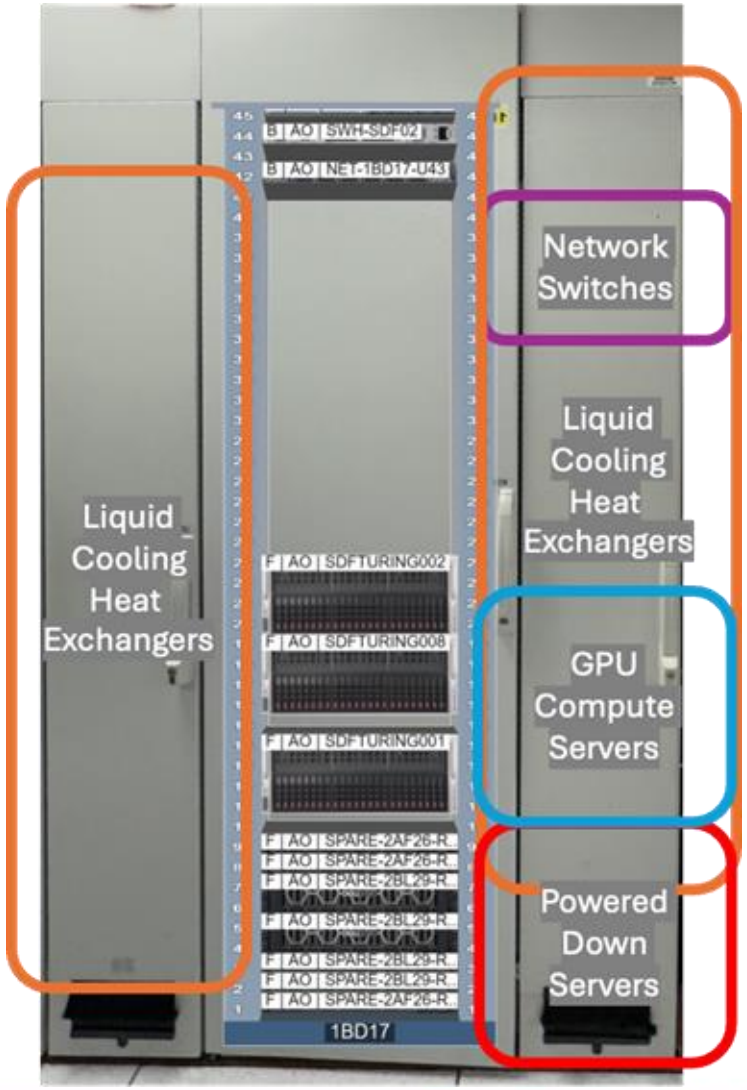
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Manual Measurement:
Lenovo SR675 – Temperature Heatmap (607 W)



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SLAC IT Data Center: Layout of B50 Server Rack





Server Rack in B50

- Placed between two heat liquid cooled heat exchangers
- Top contains networking hardware
- GPU Servers located near middle
- Powered down servers at bottom

LCP Cabinet allows for analysis for internal water-cooling efficiency

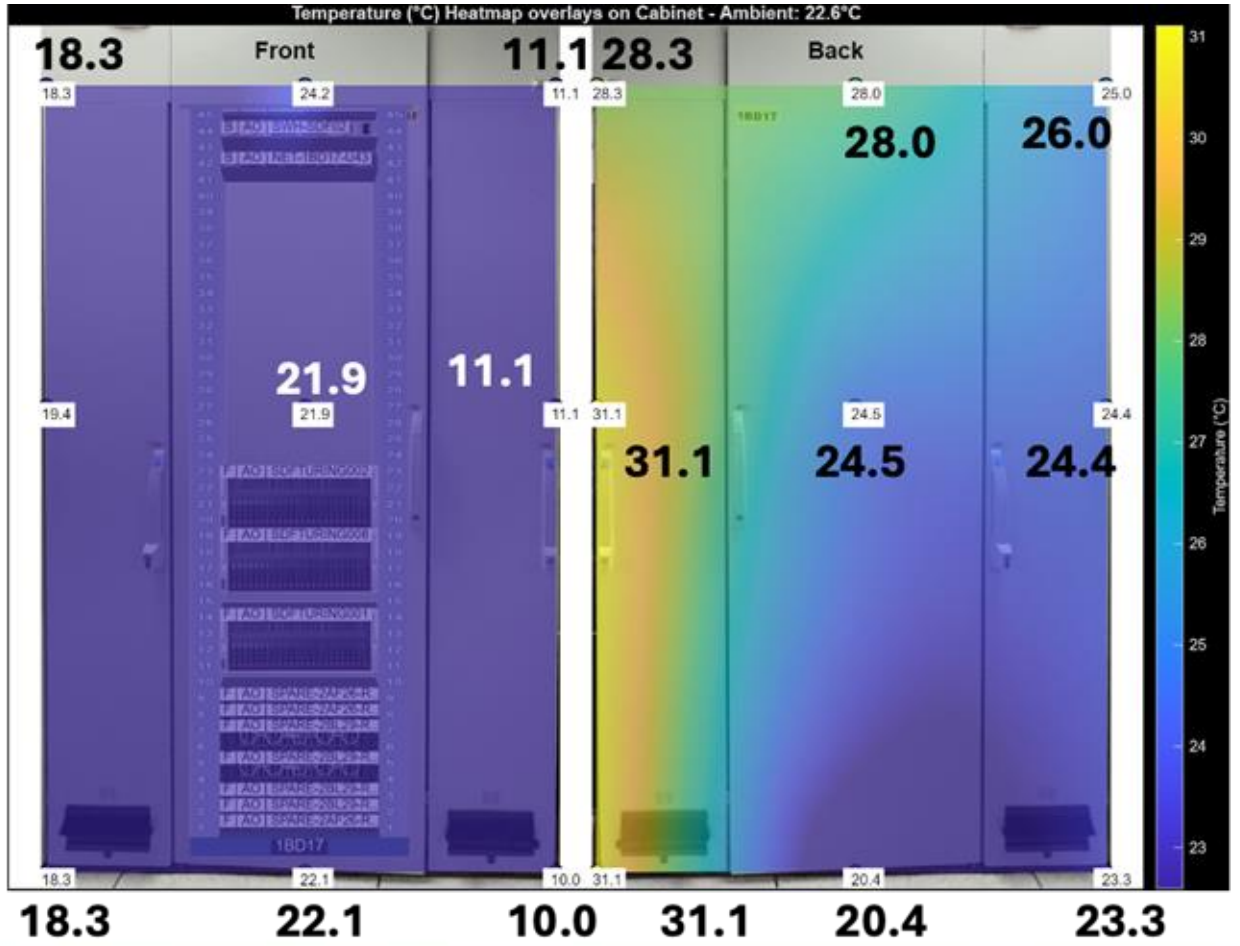
- Access to Nlyte
  - Inlet/outlet temperatures
  - Fan speeds
  - Water flow
- Access to both LCP coolers on either side of cabinet
  - Temperatures
  - Air in/out
  - Water in/out
  - Top, middle, bottom
  - Fan Speeds

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Data Center– Temperature Heatmap

- Centered temperatures were taken 4 inches from intake (**C**)
- Points at the side represent the displayed temperature of the LCP on the cabinets (left and right) themselves
- High temperature differences indicate effective cooling, or high surface area radiators

19.4



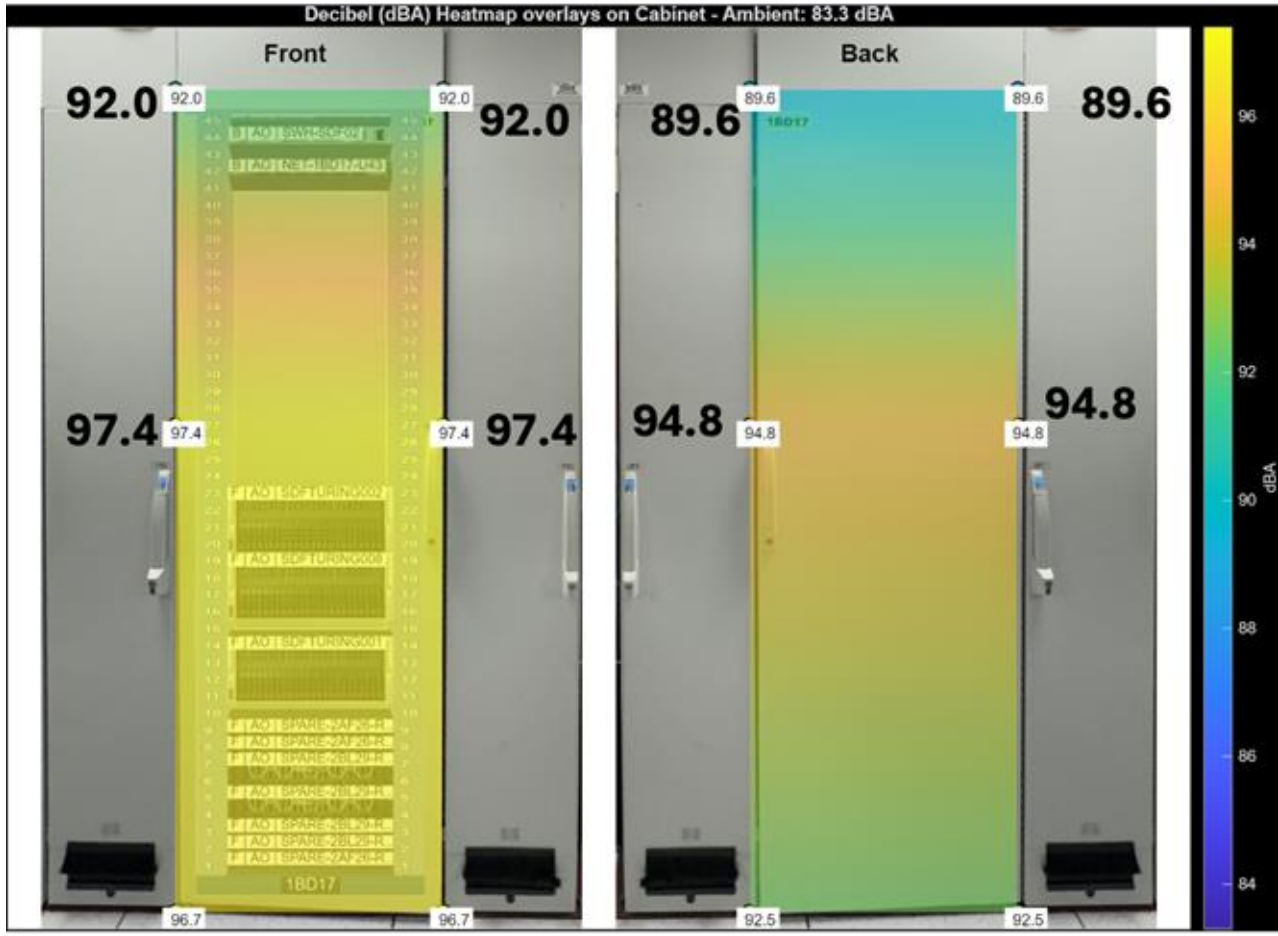
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Data Center– Sound Heatmap

- **The top of the rack is populated with network hardware**
  - Low fan noise
- **The center contains running GPU clusters**
  - Highest fan noise 92-97.4 dB
- **The bottom contains powered down servers**
  - Proximity to GPU clusters as well as ground amplifies dBA value
- **Sound levels higher than in Aircraft Cabins** *(~88 dB measured by CeecS team)*
  - In 2018, a study of sound levels measured on 200 flights representing six aircraft groups found media noise level of 83.5 db(A) with levels reaching 110 dB(A) on certain flights, but only 4.5% exceeded the NIOSH recommended 8-hr TWA of 85 dB(A)

(https://en.wikipedia.org/wiki/Aircraft_noise_pollution)



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Data Center Server Fan Profiling – Data Processing

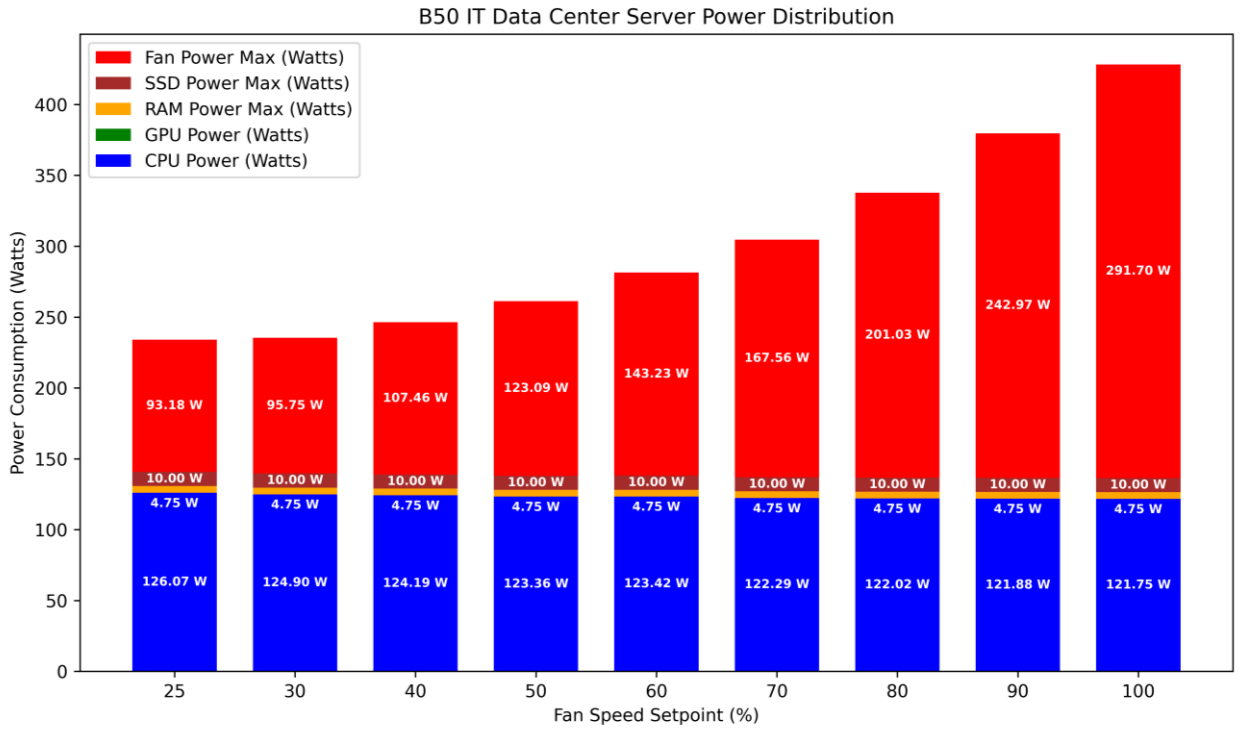
- Fan power can be derived from total system power and the power of its subcomponents when a system is idle by varying fan speeds
  - Non-pollable components like SSDs and I/O systems can be estimated from manufacturer specifications
  - RAM Power is also estimated due to a low update frequency

Fan Power Equation

$$P_{Fan} = P_{System} - (P_{CPUs} + P_{GPUs} + P_{RAM} + P_{Storage})$$

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Using Software Tools: Estimated Fan Profiles



B50 IT Data Center Server(Dell PowerEdge R6525)

- Devices
  - 2x AMD EPYC CPUs (Measured)
  - 128 GB DDR4 RAM (Estimated)
  - 2x 1TB SSD (Estimated)
  - 8x High-Performance dual Fan Modules (Estimated)
- Server fan speeds were ramped up in discrete steps while **software tools** were used to poll power from the server BMC

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Data Center Efficiency Metrics (DCEM)- workstream update

- DCEM workstream members delivered the following areas of work this year
  - Case study of metrics measurements; next steps for improvements - on going
  - Geographics PUE (gPUE) v0.1 – White Paper
  - High Density Comptue Capacity (HDCC) v0.2 – White Paper
  - Infrastructure Utilization Efficiency (IUE) v0.1.0 draft
- Plan for 2027
  - Complete gPUE, HDCC, IUE work; turn the study into actionable metrics usable in DC
- DCEM Workstream
  - Meets alternate Friday's 8:00 am to 9:00 am PT
  - https://www.opencompute.org/community/sustainability

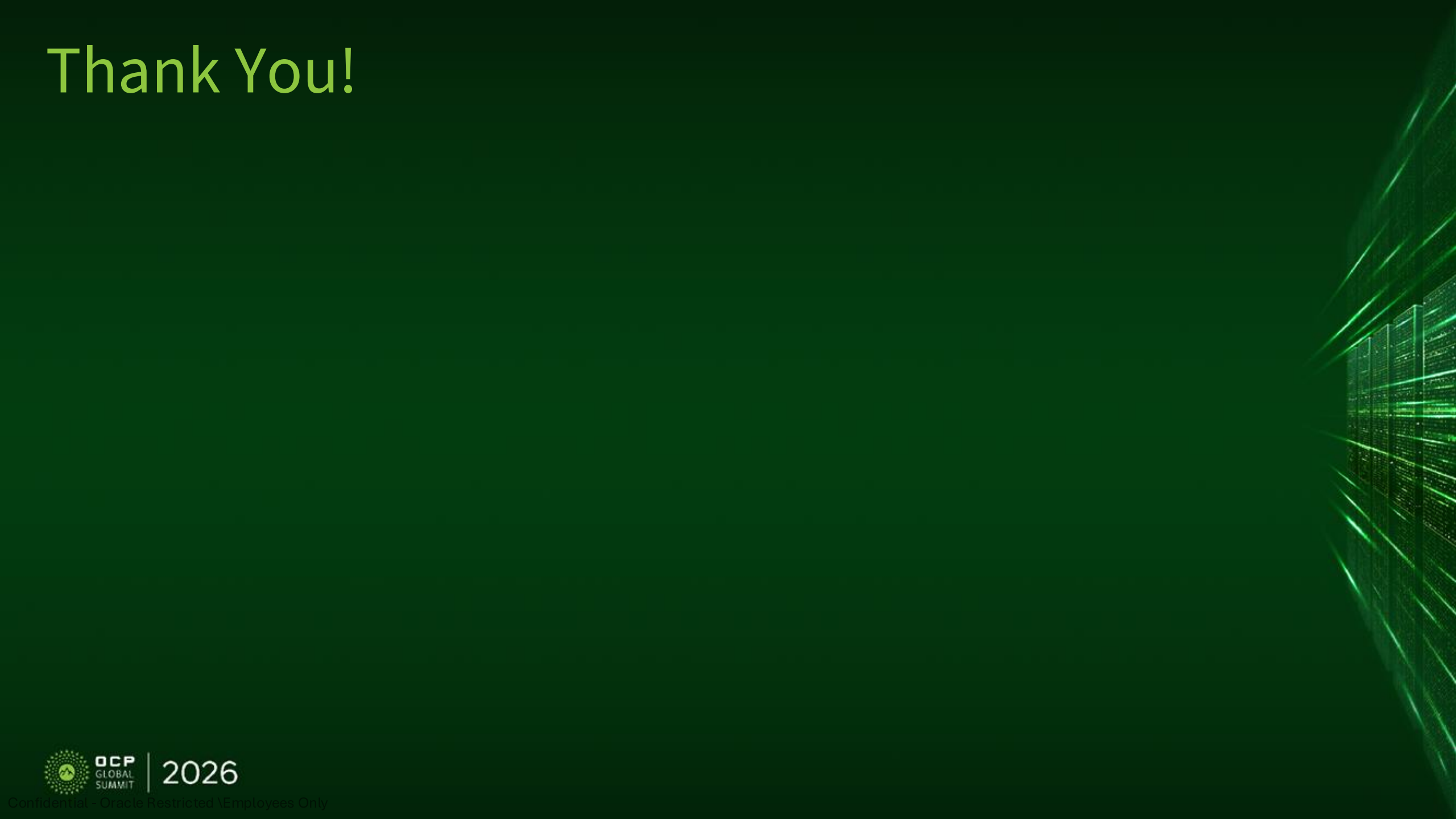
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Call to Action

- DC energy efficiency metrics plays a critical role in continuous improvement in energy efficiency
- Work with DCEM workstream to complete the definition of measurable DC energy efficiency metrics and adapt in DC
  - Complete the gPUE, HDCC white papers and work towards specification
  - Complete the IUE specification and the OCP profile definition
  - Turn the study into actionable metrics usable in DC
- https://www.opencompute.org/community/sustainability

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Thank You!



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